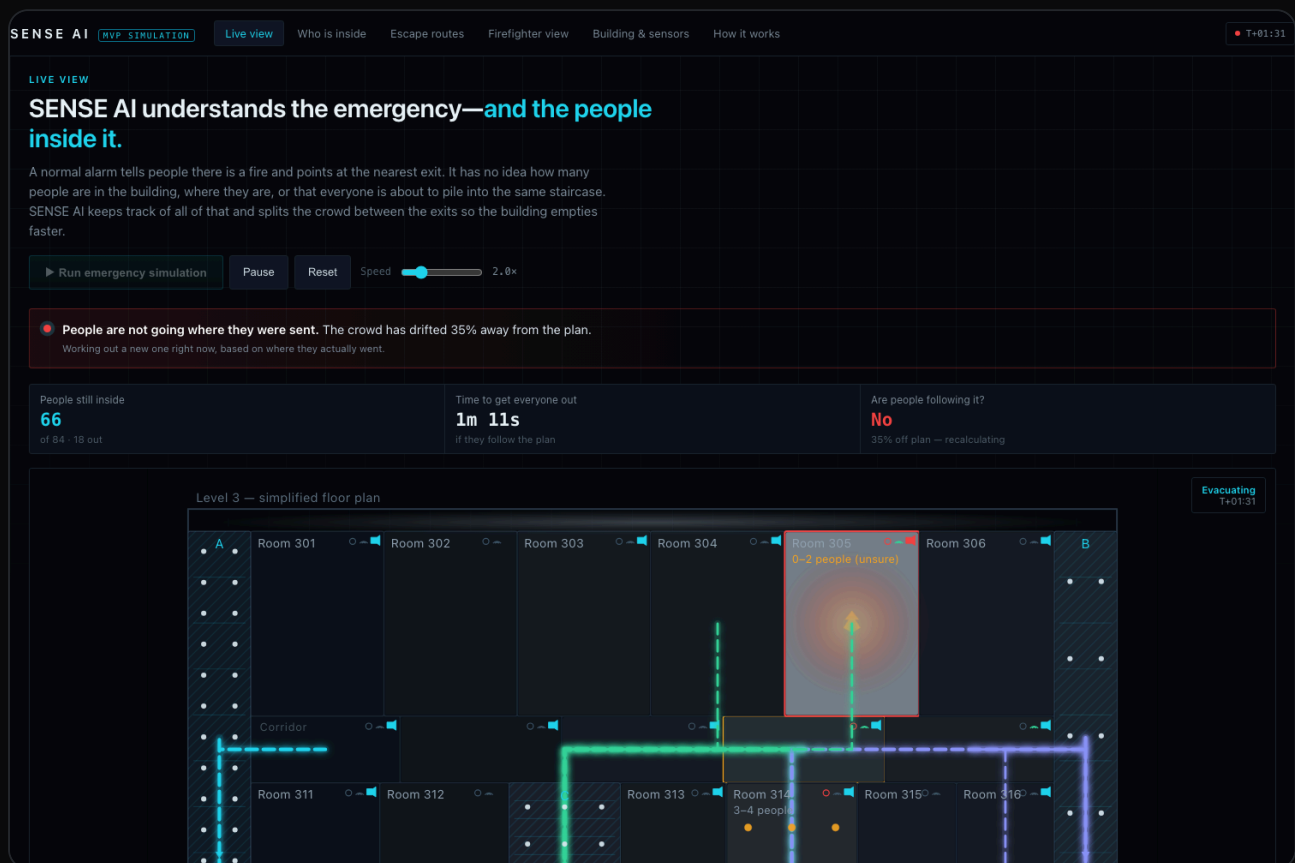


SENSE AI — what the product does

Six pages. This is our own reference, so we can answer anything a judge asks about the software. Run it with: `.venv/bin/python -m uvicorn app.main:app --port 8010` then open **127.0.0.1:8010**.

1 · Live view — the whole story in one screen



Press **Run emergency simulation**. Everything else on this page follows from it.

The sentence at the top

Plain English, changes as the fire does. It is the only thing a non-technical viewer needs to read.

Three numbers

People still inside · time to get everyone out
· whether people are following the plan.

Tabs under the map: how busy each stair is, what people are being told through the speakers, and the event log. **Speed slider** runs the incident faster for a demo.

The whole incident is 14 scripted beats and takes about 3 minutes 40 seconds at normal speed. 84 people, every one accounted for at the end.

2 · Who is inside — counting people

SENSE AI

MVP SIMULATION

Live view

Who is inside

Escape routes

Firefighter view

Building & sensors

How it works

T+01:47

WHO IS INSIDE

Know where people are.

A camera counts people well — until smoke fills the room, and then it sees nothing. A radar sensor sees straight through smoke, but it can never give an exact number. So we use both, and each one tells us how sure it is. When they are not sure enough between them, we say “4 to 6 people” instead of pretending to know it is 5.

People in each room

about 10 people on this floor

Show empty rooms

Room	People	How sure we are	Who is telling us	Air
East-central corridor	0–1	Certain	Radar	Smoke
East corridor	0–1	Certain	Radar	Some haze
Room 305 plant and server room	0–1	Not sure	Radar	Thick smoke
Room 306 open-plan desks	0–1	Certain	Camera	Some haze
Room 314 open-plan desks	4–5	Fairly sure	Radar	Smoke
Room 315 quiet room	0–1	Certain	Camera	Some haze

Try breaking it

Fill the building with smoke and watch the cameras go blind. The numbers turn into ranges, and the radar takes over the counting.

How much smoke: 0%

Fill with smoke

Clear the air

Switch off three cameras

Live map

Where that number came from

click any room above, or on the map

Room 314 — open-plan desks

Our answer: 4–5 people — we are fairly sure (85%)

Sensor	Says	Sure	We use it
Camera	2.1	55%	as a minimum only
Radar	4.6	66%	In full

We are not sure enough to give one number, so we give a range. When planning the escape we assume the worst case — 5 people.

The camera here is blinded by smoke. It can only miss people, never invent them — so we use its count as a minimum and let the radar decide the real number. Averaging the two would have made us think fewer people were in here than there are.

What the camera actually sees

YOLO LIVE

Play the example

Choose file

No file chosen

Use my own video

Each room, how many people, and how sure we are.

The controls that matter in a demo

- **Fill with smoke** — drag the slider up. Camera confidence collapses, the numbers turn into ranges like “4–6”, and the source switches from Camera to Radar.
- **Switch off three cameras** — proves the radar layer alone still holds a count.
- **Click any room** — shows exactly how that number was reached: what each sensor said, how sure it was, and how it was used.

The one technical idea worth knowing

A camera blinded by smoke can only **miss** people, never invent them. So when it is obscured we use its count as a **minimum**, not an average. Averaging would drag the estimate below the truth exactly when a life depends on it.

3 · What the camera actually sees

What the camera actually sees

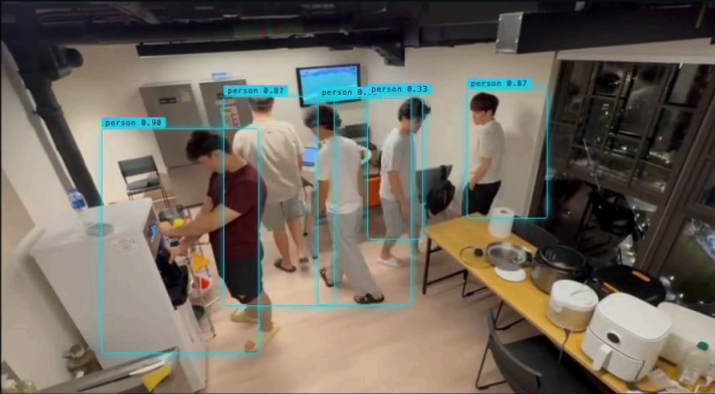
YOLO LIVE

▶ Play the example

Choose file sample.mp4

Use my own video

sample.mp4 · 40 frames sampled over 10.5s · 190 detections



People it can see
5
peak 7 in clip

How sure it is
76%
about what it found

How much smoke
0%
blocking the view

Fire or smoke?
none
spotted in the picture

Camera count usable; fused with the presence layer.

Real YOLO detection on our own uploaded footage — seven people, 84% confidence.

Play the example runs a bundled clip. **Use my own video** takes any MP4 with real people in it. The boxes are the model's actual output. Cartoons and animations will find nothing — it is trained on real people.

4 · Escape routes — break it on purpose

SENSE AI

MVP SIMULATION

Live view

Who is inside

Escape routes

Firefighter view

Building & sensors

How it works

T+02:05

ESCAPE ROUTES

The shortest route is not always the fastest.

If everyone heads for their nearest staircase, one of them jams and the other two stand half empty — and the whole building takes longer to clear. SENSE AI works out how long each route will really take once it is crowded, then splits people between all three. Press a button below to break something and watch the routes move.

Interventions

Evacuate the building

Start a fire in Room 305

Jam up Stair A

Block the east-central corridor

Start over

+ Pick a room on the map first

People are 38% off the plan — working out a new one.

observed movement diverges from the plan

Time to get everyone out

1m 31s

until the last person leaves

People given a route

10

right now

Times replanned

44

since the start

Level 3 — simplified floor plan

Room 301

Room 302

Room 303

Room 304

Room 305

Room 306

Room 311

Room 312

Room 313

Room 314

Room 315

Room 316

Corridor

Way out

Way out

Way out

Status Evacuating

T+02:05

observed movement diverges from the plan

Press the buttons on the left and watch the coloured routes move.

BUTTON	WHAT IT PROVES
Start a fire in Room 305	Routes bend away from the smoke as it spreads.
Jam up Stair A	The allocation moves people to the other two staircases.
Block the corridor	Whole sections reroute. Push it far enough and it will tell you people are trapped.
Five ready-made situations	Normal, fire near an exit, stair crowding, camera failure, two fires at once.

The tab that wins arguments: *Who goes where, and why* — every route has a written reason, like “Stair A at capacity, C is faster despite the distance”.

5 · Firefighter view

SENSE AI

MVP SIMULATION

Live view

Who is inside

Escape routes

Firefighter view

Building & sensors

How it works

T+02:21

FIREFIGHTER VIEW

See who remains—and how to reach them.

When a crew arrives they usually know almost nothing: how many people are left, or where. This screen shows which rooms still have people in them, how we know, how sure we are, and a way in that avoids fighting through the crowd coming out. It is information for the officer in charge — it does not give orders, and it is not an approved safety system.

Tower B - Evacuating

T+02:21

20 of 84 still inside

Level 5

Model: 14 sensors

Row 301

Row 302

Row 303

Row 304

Row 305

Row 306

Row 311

Row 312

Row 313

Row 314

Row 315

Row 316

Row 318

Row 319

Row 320

Row 321

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Row 1464

Row 1465

Row 146

6 · Building & sensors

SENSE AI

MVP SIMULATION

Live view

Who is inside

Escape routes

Firefighter view

Building & sensors

How it works

T+02:32

BUILDING & SENSORS

Every sensor knows which room it is in.

The system holds a model of the building: every room, corridor, staircase and door, with how wide each one is and therefore how many people can get through it per minute. Every camera and sensor is tied to a room in that model. Click any room to see what is in it.

Level 3 — simplified floor plan

Click any room R305

Room 301

Room 302

Room 303

Room 304

Room 305

Room 306

Corridor

Room 311

Room 312

Room 313

Room 314

Room 315

Room 316

Way out A

Way out C

Way out B

Small icons in each room: camera · radar · smoke detector

Tower B — drawing not to scale

The room you picked

ROOM

Room 305 — plant and server room

Floor 3

Floor area 75 m²

People inside 0-2 (not sure)

Ways out of this room

Leads to	Distance	Width	People per minute	Kind
East-central corridor	6 m	1.1 m	86	door

Structure

Building Tower B — Innovation Park

Floors 5

Mapped in detail Level 3, room by room

Spaces 23

Doors & passages 22 (bidirectional)

Staircases A · B · C

Ways out A · B · C

Sensors & devices 70

Only Level 3 is mapped room by room. For the other floors we only know roughly how many people are on them — so that is all we show, rather than inventing rooms nobody has surveyed.

What is installed

70 TOTAL

Device	How many	Status
Cameras that count people	14	in the demo
Radar sensors (see through smoke)	20	in the demo
Smoke detectors	17	in the demo
Heat detectors	2	in the demo
Speakers for announcements	8	in the demo
Ceiling arrows that point the way	8	not built yet
Link to the fire alarm panel	1	not built yet

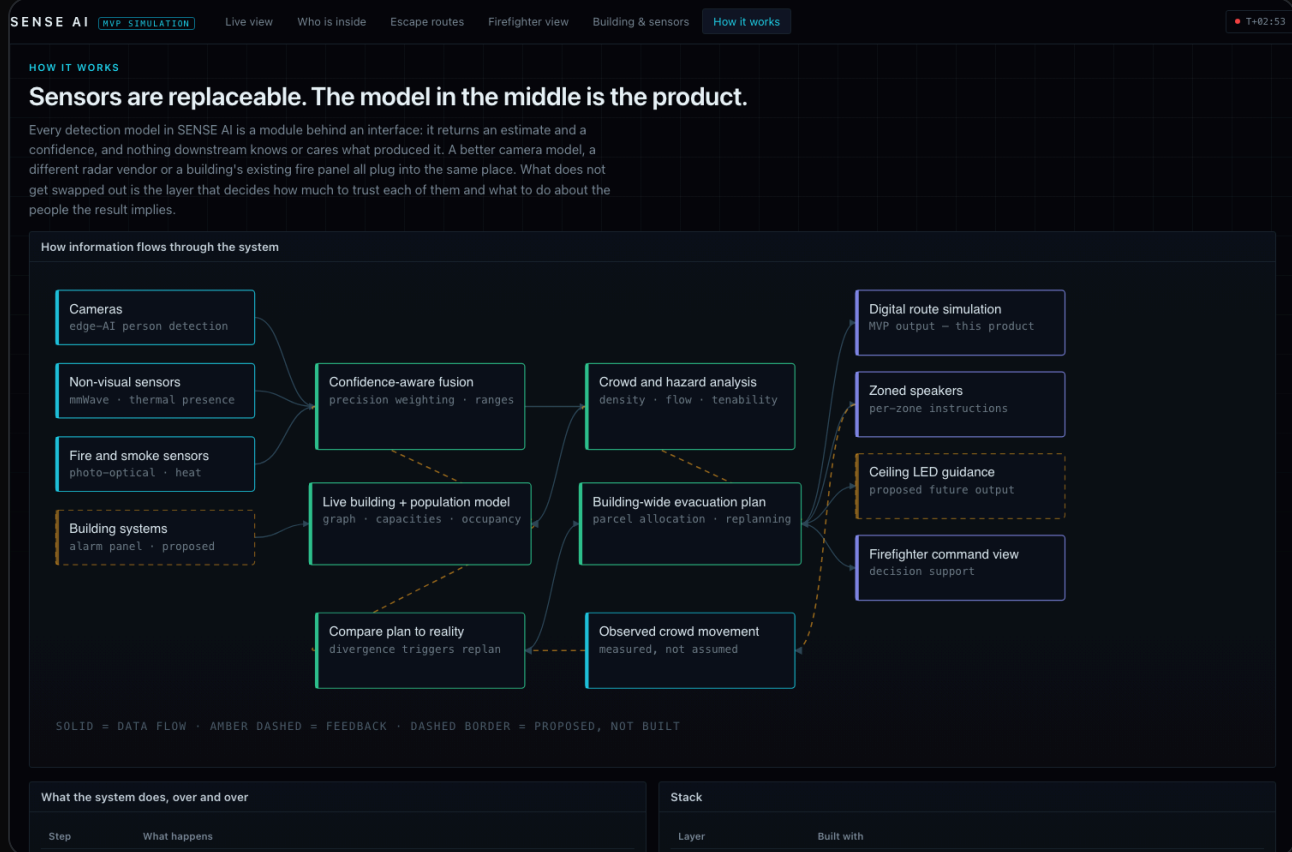
+ What exists today vs what we would build

Click any room to see what is installed in it.

The system holds a model of the building: every room, corridor and staircase, how wide each door is, and therefore how many people per minute can get through it. 23 spaces, 70 devices.

Marked “in the demo”: cameras, radar, smoke and heat detectors, speakers. **Marked “not built yet”:** ceiling arrows and the fire-panel link. We do not pretend otherwise anywhere in the product.

7 · How it works



The architecture page — for the judge who asks whether anything real is underneath.

The loop runs continuously: **sense** → **estimate** → **plan** → **guide** → **observe** → **compare** → **recalculate**. This page also lists, in plain terms, everything the prototype *cannot* do — no certification, no guaranteed detection, no seeing through walls.

If the laptop has no internet

The website needs the server running locally, so it works offline — but only on a machine with the project installed. **For the pitch, present from SENSE-AI-pitch.html instead.** The recorded clips in it are the same software, and nothing can crash mid-demo.